```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import scipy.stats as ss
from scipy.stats import stats, norm, ttest_ind, chi2_contingency, levene, ttest_ind, f_oneway
import warnings
warnings.filterwarnings('ignore')
```

Basic Data Exploration

```
# Loading the dataset
df = pd.read_csv('bike_sharing.csv')
```

```
# Verify data loaded, previwing the first 5 rows
df.head()
```

	datetime	season	holiday	workingday	weather	temp	atemp	humidity	windspeed	casual	registered	count
0	2011-01-01 00:00:00	1	0	0	1	9.84	14.395	81	0.0	3	13	16
1	2011-01-01 01:00:00	1	0	0	1	9.02	13.635	80	0.0	8	32	40
2	2011-01-01 02:00:00	1	0	0	1	9.02	13.635	80	0.0	5	27	32
3	2011-01-01 03:00:00	1	0	0	1	9.84	14.395	75	0.0	3	10	13
4	2011-01-01 04:00:00	1	0	0	1	9.84	14.395	75	0.0	0	1	1

```
# Check the number of rows and columns of the dataframe
df.shape
```

```
(10886, 12)
```

```
# Check the dataframe schema and missing values
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10886 entries, 0 to 10885
Data columns (total 12 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   datetime    10886 non-null  object
 1   season      10886 non-null  int64
 2   holiday     10886 non-null  int64
 3   workingday  10886 non-null  int64
 4   weather     10886 non-null  int64
 5   temp        10886 non-null  float64
 6   atemp       10886 non-null  float64
 7   humidity    10886 non-null  int64
 8   windspeed   10886 non-null  float64
 9   casual      10886 non-null  int64
 10  registered  10886 non-null  int64
 11  count       10886 non-null  int64
dtypes: float64(3), int64(8), object(1)
memory usage: 1020.7+ KB
```

```
df['datetime'] = pd.to_datetime(df['datetime'])
```

```
df[['season', 'holiday', 'workingday', 'weather']] = df[['season', 'holiday', 'workingday', 'weather']].astype('category')
```

```
# Check basic stats of the numerical columns
df.describe()
```

	datetime	temp	atemp	humidity	windspeed	casual	registered	count
count	10886	10886.00000	10886.000000	10886.000000	10886.000000	10886.000000	10886.000000	10886.000000
min	2011-01-01 00:00:00	0.82000	0.760000	0.000000	0.000000	0.000000	0.000000	1.000000
50%	2012-01-01 20:30:00	20.50000	24.240000	62.000000	12.998000	17.000000	118.000000	145.000000
max	2012-12-19 23:00:00	41.00000	45.455000	100.000000	56.996900	367.000000	886.000000	977.000000

```
# Check the category variables
df.describe(include='category')
```

	season	holiday	workingday	weather
count	10886	10886	10886	10886
unique	4	2	2	4
top	4	0	1	1
freq	2734	10575	7412	7192

```
# Check for duplicate values in the df
df.duplicated().sum()
```

```
np.int64(0)
```

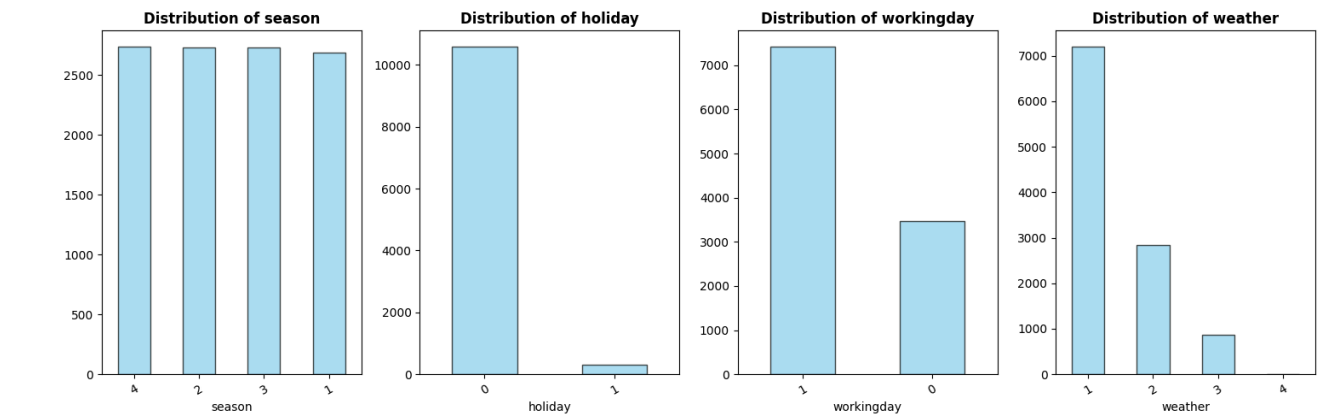
Inference:

- The dataset contains 10,886 rows and 12 columns (features).
- No missing values are present across any of the columns.
- The datetime column is currently stored as an object, and the season, holiday, workingday, and weather columns are categorical variables stored as integers. These columns are converted to their appropriate data types to improve clarity and support more effective analysis.
- The dataset does not contain any duplicate data.
- The dataset contains two years of data.
- Temparature and atemp shows similar central tendency and spread suggesting a relationship.
- The humidity is ranging from 0 to 100 however the majority of the data lies between 47 to 77 suggesting humid conditions.
- Windspeed data is ranging from 0 to 57 kmph with majority of data 7 to 17 kmph range suggesting normal wind speed events, though a few high wind observations also made.
- Bike rental demand (casual, registered, and count) shows strong right-skewness, as the mean is consistently higher than the median, indicating the presence of occasional peak-demand periods.
- Registered users contribute significantly more to total rentals than casual users, both in terms of average usage and maximum values, suggesting that regular commuters form the core user base.
- Extreme difference between the 75th percentile and the max value of the casual, registered, and count columns indicates extreme usage spikes, probably driven by special events.

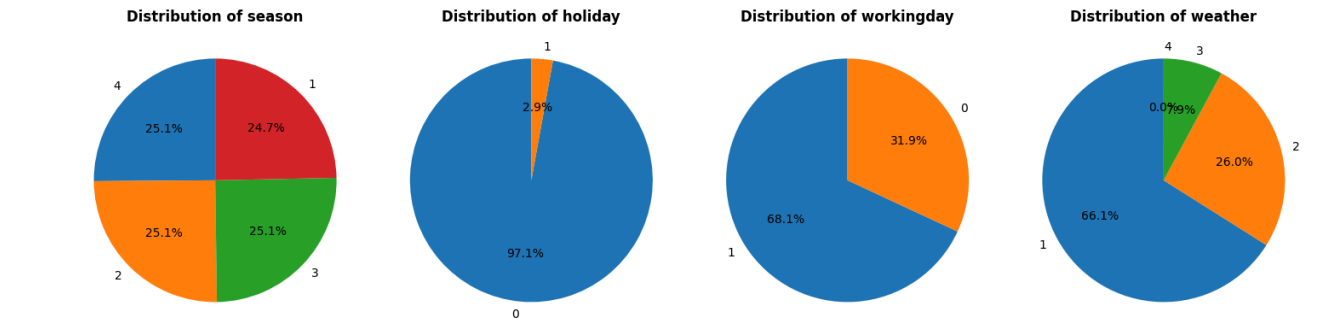
Univariate Analysis - Visualization

```
# Category column extraction for visualization
cat_cols = ['season', 'holiday', 'workingday', 'weather']
```

```
fig, axes = plt.subplots(1, len(cat_cols), figsize=(15, 5))
for ax, col in zip(axes, cat_cols):
  df[col].value_counts().plot(
      kind='bar',
      ax=ax,
      edgecolor='black',
      color='skyblue',
      alpha=0.7

  )
  ax.set_title(f'Distribution of {col}', fontsize=12, fontweight='bold')
  ax.tick_params(axis='x', rotation=30)

plt.tight_layout()
plt.show()
```




```
fig, axes = plt.subplots(1, len(cat_cols), figsize=(15, 5))

for ax, col in zip(axes, cat_cols):
    df[col].value_counts().plot(
        kind='pie',
        ax=ax,
        autopct='%1.1f%%',
        startangle=90
    )
    ax.set_title(f'Distribution of {col}', fontsize=12, fontweight='bold'),
    ax.set_ylabel('')

plt.tight_layout()
plt.show()
```



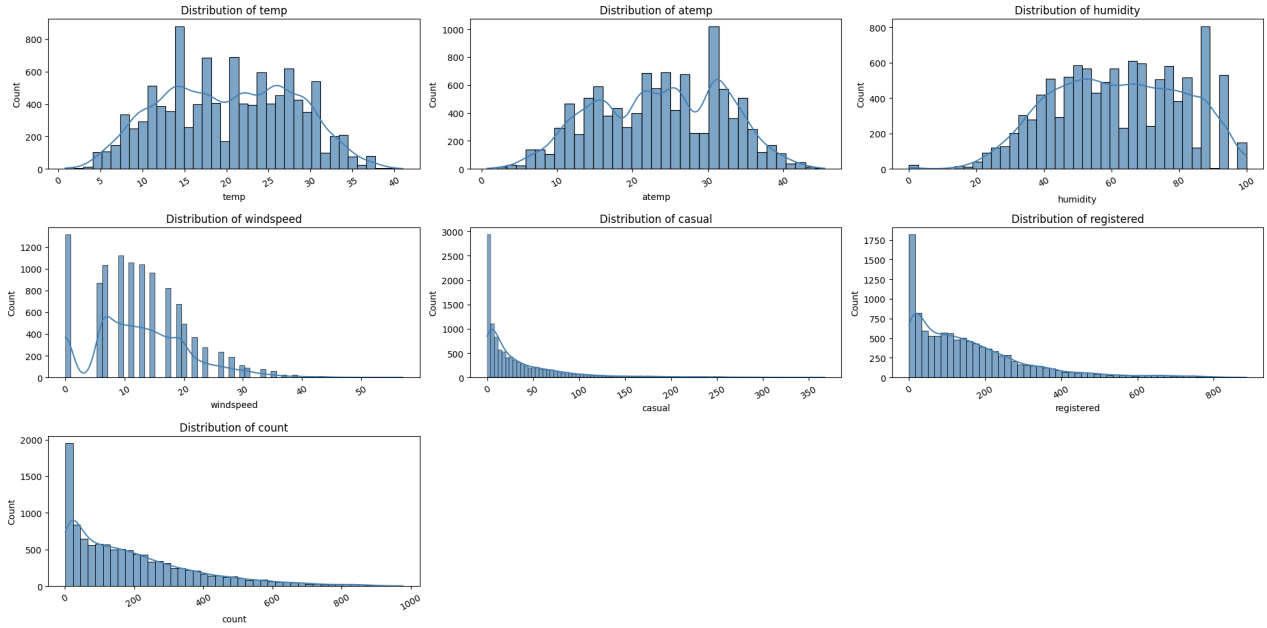
Inference:

- The seasonal distribution is fairly uniform across all four seasons, indicating that bike usage is consistently spread throughout the year with no strong seasonal dominance.
- Bike usage on holidays is minimal, suggesting that rentals are significantly lower on holidays compared to non-holidays.
- A majority of the observations (68 %) correspond to working days, indicating that bike usage is higher on working days, likely driven by commuting behavior.
- Bike usage is highest under clear weather conditions, followed by misty and light snow conditions, suggesting that users strongly prefer favorable weather for riding.
- Usage is negligible during severe weather conditions (heavy rain, snow, or storms), indicating that adverse weather significantly discourages bike rentals.

```
# Numerical column extraction for visualization.
num_cols = ['temp','atemp', 'humidity', 'windspeed', 'casual', 'registered', 'count']
```

```
fig = plt.figure(figsize=(20, 10))

for i, col in enumerate(num_cols, 1):
    ax = fig.add_subplot(3, 3, i)
    sns.histplot(
        data=df,
        x=col,
        kde=True,
        edgecolor='black',
```

```
        color='steelblue',
        alpha=0.7,
        ax=ax
    )
    ax.set_title(f'Distribution of {col}')
    ax.tick_params(axis='x', rotation=30)

plt.tight_layout()
plt.show()
```



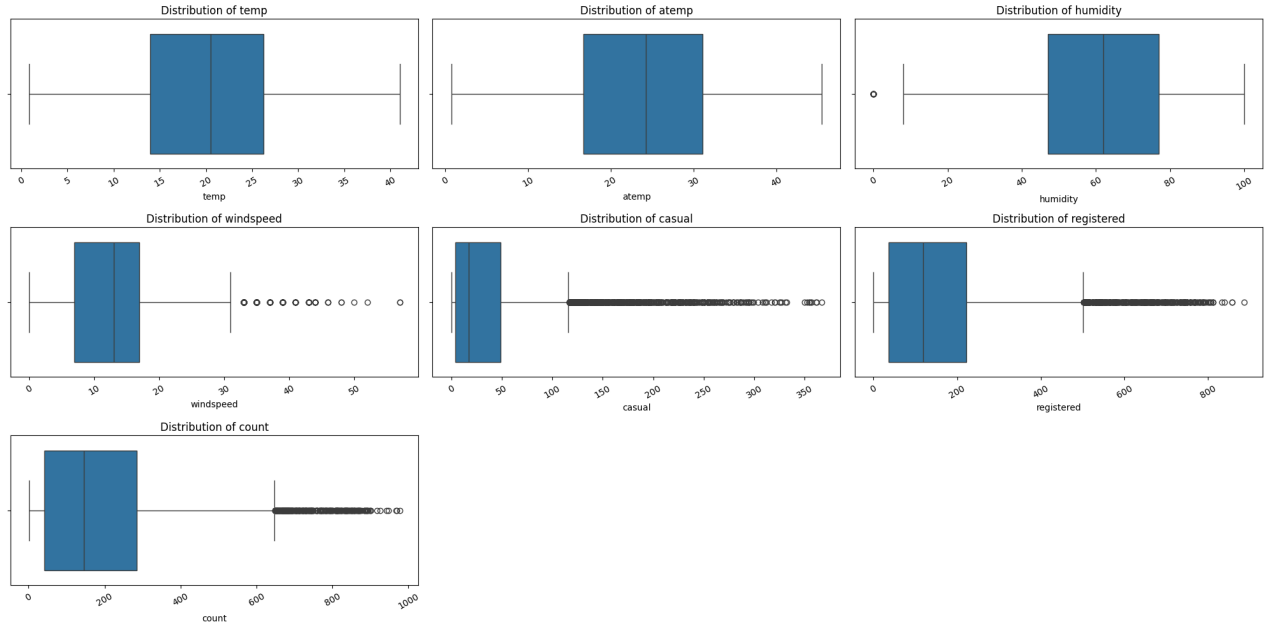
Inference:

- The temperature column distribution is normal without any skewness with most of the values lying between 10 to 30 derees celcius suggesting that the users prefer moderate temperature.
- The actual temperature column is closely resumbling the temperature column suggesting high correlation and the values mostly lie between 10 to 35 degrees celcius.
- The humidity distribution is slighlty left skewed suggesting that the humidity is high.
- The windspeed is right skewed with most wind speed between 0 to 20 kmph.
- The Casual usage column is highly right-skewed distribution with large number of days with low casual users and few days with very high casual usage suggesting casual usage is inconsistent and event-driven, Strongly affected by weather, weekends, or holidays.
- The registered users uage distribution is right skewed, but less extreme than casual users, more stable and spread out distribution. The usage is higher compared to casual users. Registered users are less sensitive to random external factors indicates habitual usage like commuting, daily routines.

Outlier detection and handling

```
fig = plt.figure(figsize=(20, 10))

for i, col in enumerate(num_cols, 1):
    ax = fig.add_subplot(3, 3, i)
    sns.boxplot(
        data=df,
        x=col,
        ax=ax
```

```
    )
    ax.set_title(f'Distribution of {col}')
    ax.tick_params(axis='x', rotation=30)

plt.tight_layout()
plt.show()
```



Inference:

- The temp and atemp columns are approximately symmetrically distributed with no extreme outliers. Since both variables are highly correlated, one of them can be dropped to reduce multicollinearity.
- The humidity distribution is mainly concentrated between 50 and 80, with a few values near the boundaries. As humidity is naturally bounded between 0 and 100, no outlier treatment is required.
- Windspeed exhibits a right-skewed distribution with a long tail. The higher values appear to be valid observations rather than anomalies, Hence no treatment is applied
- The casual variable is highly right-skewed, with occasional large spikes likely corresponding to weekends and holidays. These represent genuine demand surges rather than data errors; therefore, no outlier removal is applied.
- The registered user count shows right skewness, with higher values during peak demand periods. These observations are valid and should be retained, as they carry important business information.

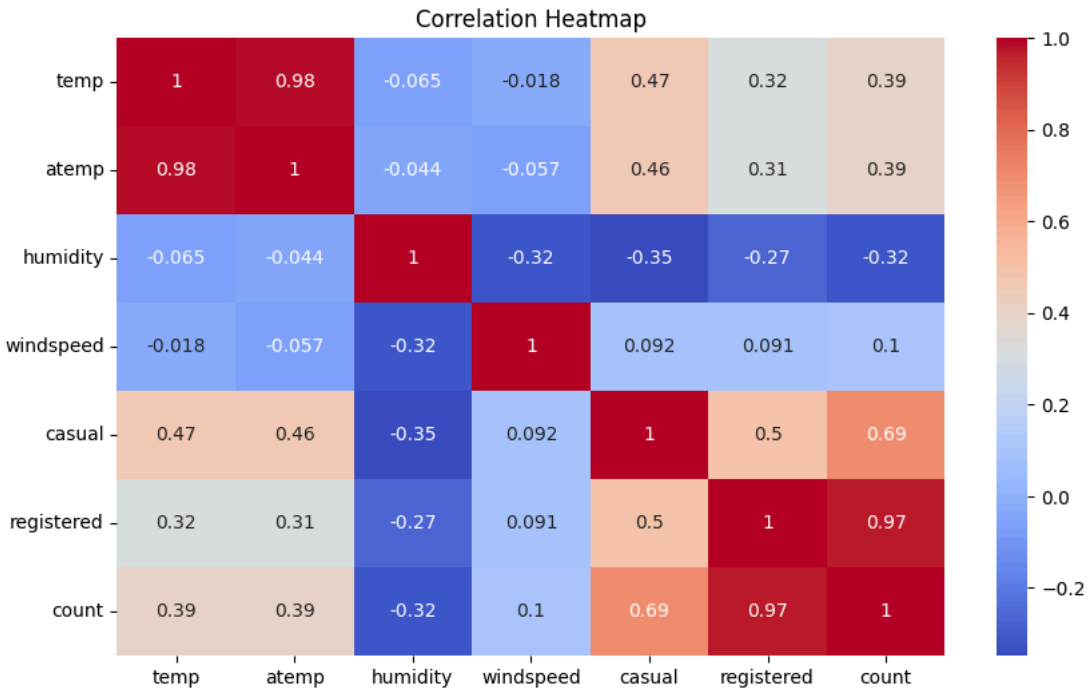
Correlation Analysis

```
num_corr_matrix = df[num_cols].corr()
```

```
num_corr_matrix
```

	temp	atemp	humidity	windspeed	casual	registered	count
temp	1.000000	0.984948	-0.064949	-0.017852	0.467097	0.318571	0.394454
atemp	0.984948	1.000000	-0.043536	-0.057473	0.462067	0.314635	0.389784
humidity	-0.064949	-0.043536	1.000000	-0.318607	-0.348187	-0.265458	-0.317371
windspeed	-0.017852	-0.057473	-0.318607	1.000000	0.092276	0.091052	0.101369
casual	0.467097	0.462067	-0.348187	0.092276	1.000000	0.497250	0.690414
registered	0.318571	0.314635	-0.265458	0.091052	0.497250	1.000000	0.970948
count	0.394454	0.389784	-0.317371	0.101369	0.690414	0.970948	1.000000

```
plt.figure(figsize=(10, 6))
sns.heatmap(num_corr_matrix, annot=True, cmap="coolwarm")
plt.title('Correlation Heatmap')
plt.show()
```



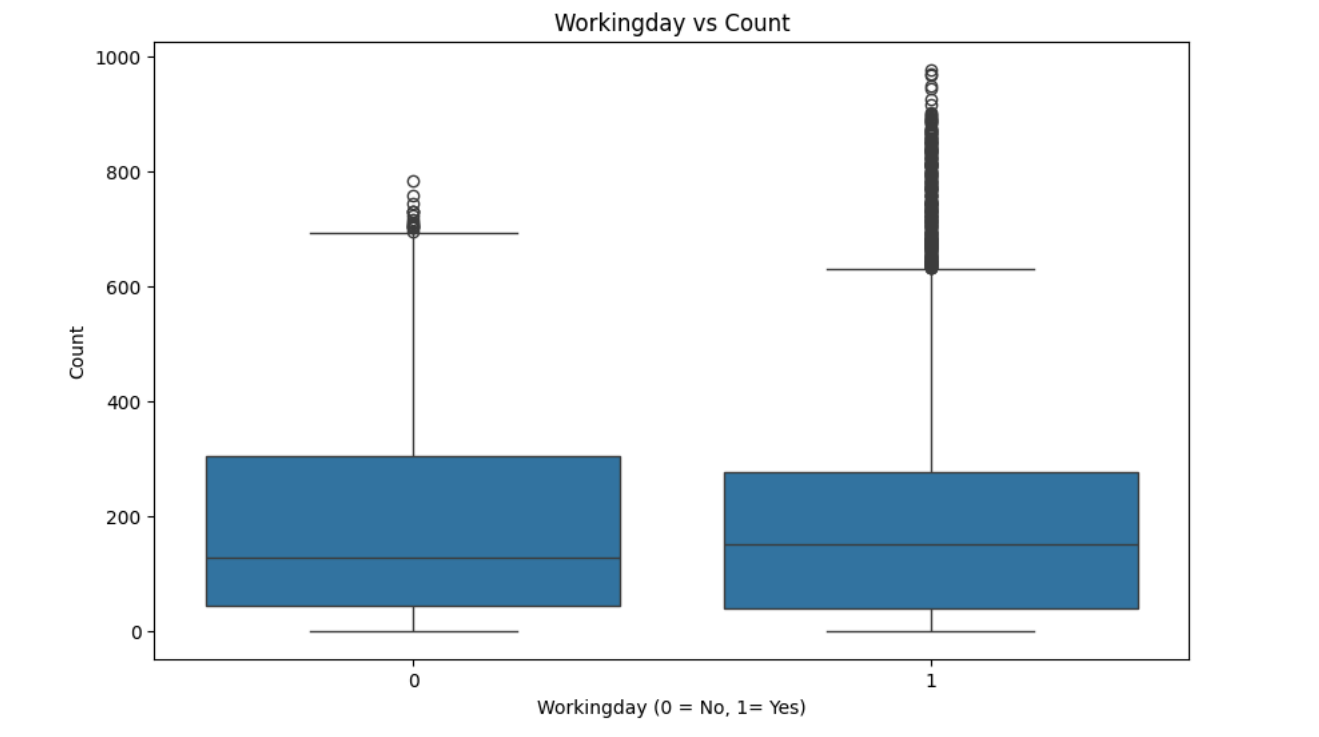
Inference:

- Registered users show a very strong positive correlation with total bike rentals, making them the primary driver of demand.
- Casual users also exhibit a strong positive correlation with total demand, especially during high-usage periods.
- Humidity has a moderate negative correlation with bike rentals, suggesting reduced demand during high-humidity conditions.
- Temperature and feeling temperature are highly correlated with each other, indicating multicollinearity, therefore, one of these variables can be dropped.

Hypothesis Testing

Workingday vs Count(Bike rentals)

```
# plot a boxplot between workingday and count

plt.figure(figsize=(10, 6))
sns.boxplot(x='workingday', y='count', data=df)
plt.xlabel('Workingday (0 = No, 1= Yes)')
plt.ylabel('Count')
plt.title('Workingday vs Count')
plt.show()
```



Inference:

- The boxplot shows a higher median number of bike rentals on working days compared to non-working days.
- Although several high-value observations are flagged as outliers, these represent genuine high-demand periods and are retained for further analysis.

Null and Alternate Hypothesis:

- Null Hypothesis h0 : There is no difference between no of bike rides on weekdays and weekends.
- Alternate Hypothesis Ha : There is no statistically significant difference in no of bike rides on weekdays and weekends.

```
working = df[df['workingday'] == 1]['count']
non_working = df[df['workingday'] == 0]['count']

levene(working, non_working)
```

```
LeveneResult(statistic=np.float64(0.004972848886504472), pvalue=np.float64(0.9437823280916695))
```

```
working_day = df[df['workingday'] == 1]['count']
non_working_day = df[df['workingday'] == 0]['count']
```

Null and Alternate Hypothesis - Levene's test:

- Null Hypothesis (H0) : The variances of count(bike rentals) are equal for working and non working days.
- Alternate Hypothesis (Ha) : The variances of count (bike rentals) are not equal for working and non working days.

```
stats, p_value = levene(working_day, non_working_day)
print('p-value:', p_value, 'and', 'stats:', stats)
```

```
p-value: 0.9437823280916695 and stats: 0.004972848886504472
```

```
if p_value < 0.05:
    print("Reject the null hypothesis. The variances are not equal.")
else:
    print("Fail to reject the null hypothesis. The variances are equal")
```

```
Fail to reject the null hypothesis. The variances are equal
```

Inference:

- Levene's test indicates no significant difference in variances between working and non-working days, confirming that the equal variance assumption is satisfied.
- As the analysis involves comparing the mean of a continuous variable (total bike rentals) across two independent groups, an independent two-sample t-test is used.

```
alpha = 0.05
stats, p_value = ttest_ind(working_day, non_working_day, equal_var=True)
print('p-value:', p_value, 'and', 'stats:', stats)
```

```
p-value: 0.22644804226361348 and stats: 1.2096277376026694
```

```
if p_value < 0.05:
  print("Reject the null hypothesis. There is a difference in no of bike rides on weekdays and weekends.")
else:
  print("Fail to reject the null hypothesis. There is no statistically significant difference in no of bike rides on weekda
```

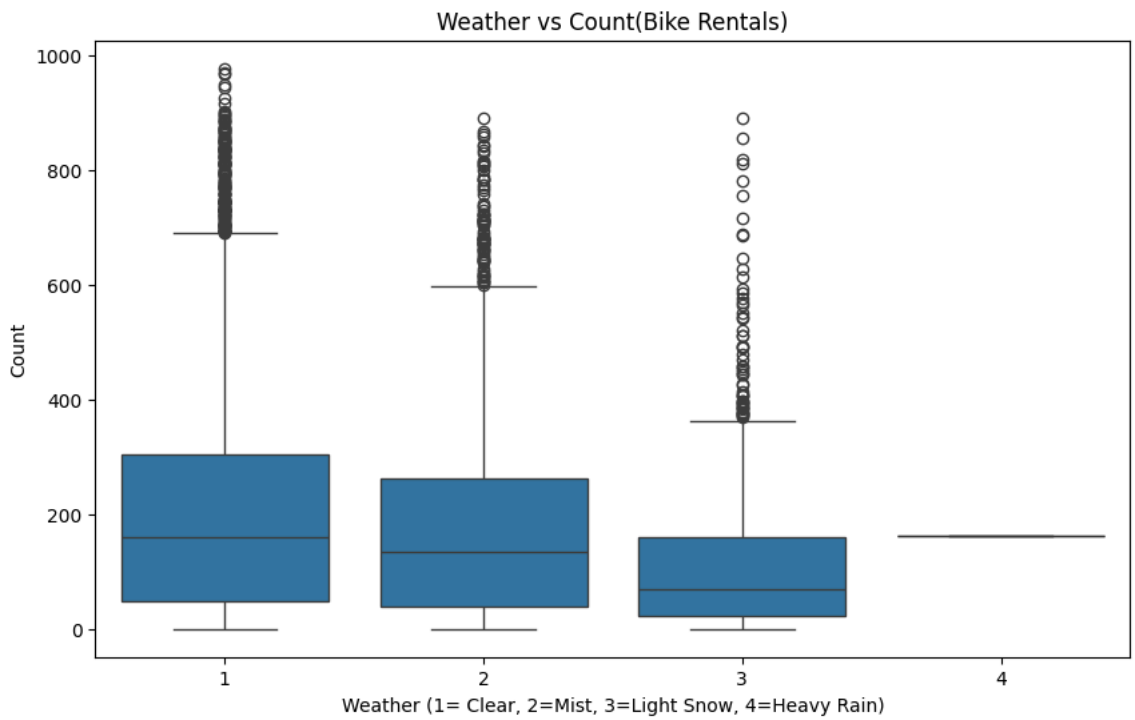
```
Fail to reject the null hypothesis. There is no statistically significant difference in no of bike rides on weekdays and wee
```

Insight:

- Since the p-value obtained from the independent two-sample t-test is greater than the significance level (α = 0.05), we fail to reject the null hypothesis. This indicates that there is no statistically significant difference in the number of bike rentals between working days and non-working days.

Weather conditions vs Count (Bike rentals)

```
# Plot a box plot between weather and count
plt.figure(figsize=(10, 6))
sns.boxplot(x='weather', y='count', data=df)
plt.xlabel('Weather (1= Clear, 2=Mist, 3=Light Snow, 4=Heavy Rain)')
plt.ylabel('Count')
plt.title('Weather vs Count(Bike Rentals)')
plt.show()
```



Insight:

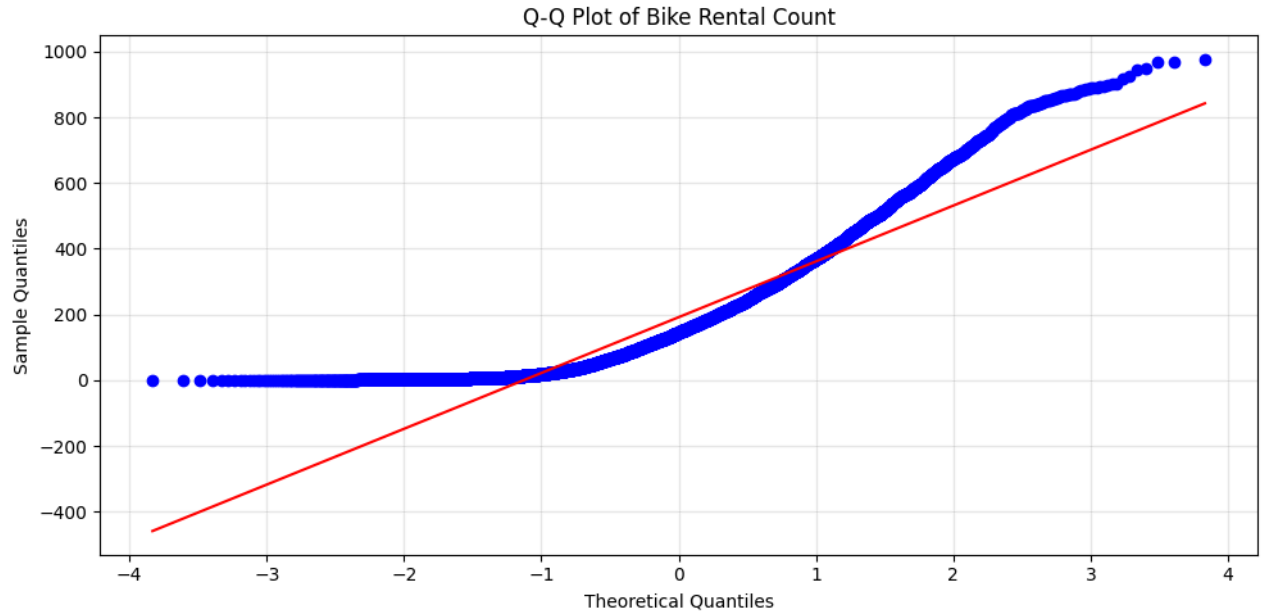
- The boxplot indicates visible variation in bike rental demand across different weather conditions, with higher demand during clear weather and lower demand during adverse weather.

- Since the dependent variable count(Bike rentals) is continuous and the independent variable (weather) is categorical with more than two levels, a one way ANOVA is an appropriate statistical test to compare the mean bike rental demand across different weather conditions.

Null and Alternate Hypothesis:

- Null Hypothesis (H0): The mean number of bike rentals is the same across all weather conditions.
- Alternate Hypothesis (Ha): The mean number of bike rentals is different on atleast one weather condition.

```
# QQ plot to check the normality of the data.
plt.figure(figsize=(10, 5))
ss.probplot(df['count'], dist='norm', plot=plt)
plt.xlabel("Theoretical Quantiles")
plt.ylabel("Sample Quantiles")
plt.title("Q-Q Plot of Bike Rental Count")
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```



Insight:

- The Q−Q plot indicates deviations from normality, particularly in the lower and upper tails, suggesting a right-skewed distribution of bike rental counts. However, given the large sample size and the robustness of ANOVA to moderate deviations from normality, the assumption is considered reasonably satisfied.

Null and Alternate Hypothesis - Levene's test:

- Null Hypothesis (H0): The variance of bike rental demand (count) is equal across all weather conditions.
- Alternate Hypothesis (Ha): The variance of bike rental demand (count) is not equal across weather conditions.

```
# Splitting Weather data into groups.
weather_1 = df[df['weather'] == 1]['count']
weather_2 = df[df['weather'] == 2]['count']
weather_3 = df[df['weather'] == 3]['count']
weather_4 = df[df['weather'] == 4]['count']
```

```
# Performing Levene's test to understand the variance accross the weather conditions.
stats, p_value = levene(weather_1, weather_2, weather_3, weather_4)
print('p-value:', p_value, 'and', 'stats:', stats)
```

```
p-value: 3.504937946833238e-35 and stats: 54.85106195954556
```

```
alpha = 0.05
if p_value < alpha:
  print("Reject the null hypothesis. The variances are not equal.")
else:
  print("Fail to reject the null hypothesis. The variances are equal")
```

```
Reject the null hypothesis. The variances are not equal.
```

Insight:

- Since the p-value obtained from Levene's test is less than the significance level (α = 0.05), we reject the null hypothesis and conclude that the variances of bike rental demand are not equal across different weather conditions.
- Given the large sample size and the robustness of one-way ANOVA to moderate violations of normality and homogeneity of variance, the analysis proceeds with one-way ANOVA.

```
# Performing one way Anova.
stats, p_value = f_oneway(weather_1, weather_2, weather_3, weather_4)
print('stats: ', stats, 'p_value:', p_value)
```

```
stats:  65.53024112793265 p_value: 5.482069475935669e-42
```

```
if p_value < alpha:
  print("Reject the null hypothesis. The mean number of bike rentals is different on atleast one weather condition.")
else:
  print("Fail to reject the null hypothesis. The mean number of bike rentals is the same across all weather conditions.")
```

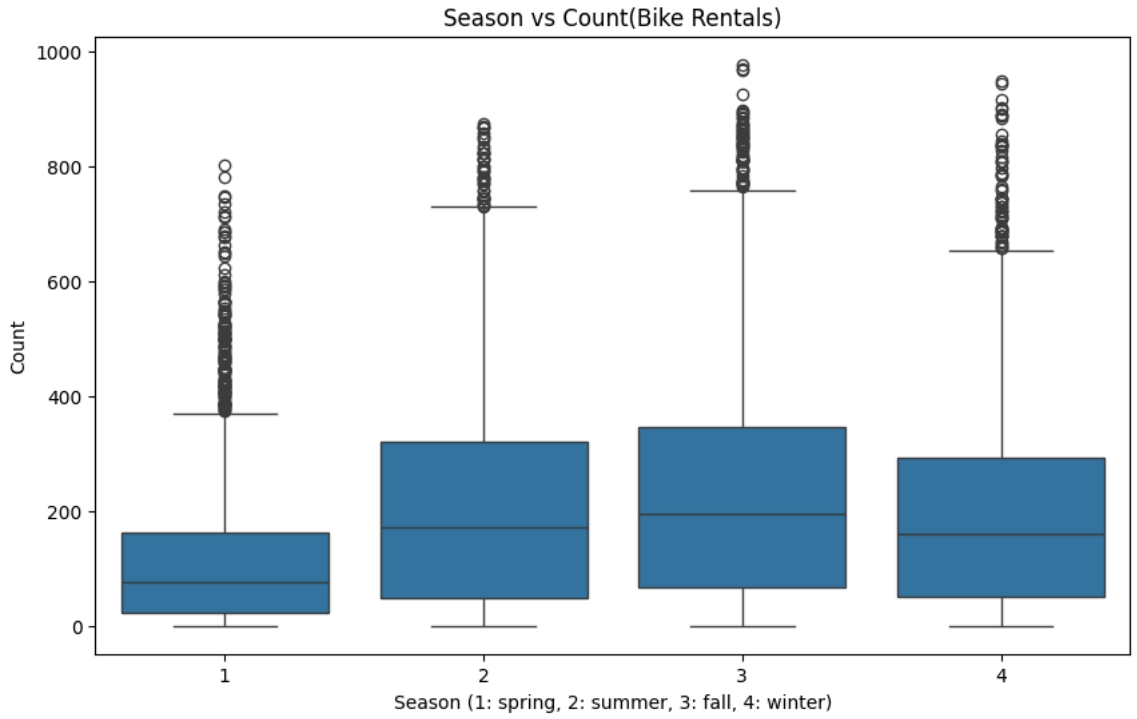
```
Reject the null hypothesis. The mean number of bike rentals is different on atleast one weather condition.
```

Inference:

- Since the p-value obtained from the one-way ANOVA test is less than the significance level (α = 0.05), we reject the null hypothesis.
- This indicates that the mean number of bike rentals differs significantly across weather conditions, with at least one weather category having a different average demand.

Season vs Count(Bike Rentals)

```
plt.figure(figsize=(10, 6))
sns.boxplot(x='season', y='count', data=df)
plt.xlabel('Season (1: spring, 2: summer, 3: fall, 4: winter)')
plt.ylabel('Count')
plt.title('Season vs Count(Bike Rentals)')
plt.show()
```



Insight:

- The boxplot indicates visible variation in bike rental demand across different seasons. Summer and fall exhibit the highest median demand with similar rental patterns, followed by winter, while spring shows the lowest overall demand. This suggests that seasonal factors influence bike rental usage.
- Since the dependent variable (bike rental count) is continuous and the independent variable (season) is categorical with more than two levels, a one-way ANOVA is an appropriate statistical test to compare the mean bike rental demand across different seasons.

Null and Alternate Hypothesis:

- Null Hypothesis (H0): The mean number of bike rentals is the same across all seasons.
- Alternate Hypothesis (Ha): The mean number of bike rentals is different on atleast one season.

Null and Alternate Hypothesis - Levene's Test:

- Null Hypothesis (H0): The variance of bike rental demand (count) is equal across all seasons.
- Alternate Hypothesis (Ha): The variance of bike rental demand (count) is not equal across seasons.

```
# Splitting the season into groups
season_1 = df[df['season'] == 1]['count']
season_2 = df[df['season'] == 2]['count']
seacon_3 = df[df['season'] == 3]['count']
season_4 = df[df['season'] == 4]['count']
```

```
# Performing levene's test variace
stats, p_value = levene(season_1, season_2, seacon_3, season_4)
print('Stats: ', stats, 'P_value: ', p_value)
```

```
Stats:  187.7706624026276 P_value:  1.0147116860043298e-118
```

```
if p_value < alpha:
  print("Reject the null hypothesis. The variances are not equal for all the seasons.")
else:
  print("Fail to reject the null hypothesis. The variances are equal for all the seasons")
```

```
Reject the null hypothesis. The variances are not equal for all the seasons.
```

Insight:

- Since the p-value obtained from Levene's test is less than the significance level (α = 0.05), we reject the null hypothesis and conclude that the variances of bike rental demand are not equal across different seasons.
- Given the large sample size and the robustness of one-way ANOVA to moderate violations of normality and homogeneity of variance, the analysis proceeds with one-way ANOVA.

```
# Performing one way Anova.
stats, p_value = f_oneway(season_1, season_2, seacon_3, season_4)
print('stats: ', stats, 'p_value:', p_value)
```

```
stats:  236.94671081032098 p_value: 6.164843386499654e-149
```

```
if p_value < alpha:
  print("Reject the null hypothesis. The mean number of bike rentals is different on atleast one season.")
else:
  print("Fail to Reject the null hypothesis. The mean number of bike rentals is equal across all seasons.")
```

```
Reject the null hypothesis. The mean number of bike rentals is different on atleast one season.
```

Inference:

- Since the p-value obtained from the one-way ANOVA test is less than the significance level (α = 0.05), we reject the null hypothesis.
- This indicates that the mean number of bike rentals differs significantly across seasons, with at least one weather category having a different average demand.

Weather Conditions vs Seasons

- Since both variables (weather and season) are categorical and the observations are independent, a ChiSquare test of independence is appropriate to examine whether there is a dependency between weather conditions and seasons

Null and Alternate Hypothesis:

- Null Hypothesis (H0): Weather conditions and season are independent of each other.
- Alternate Hypothesis (Ha): Weather conditions and seasons are dependent on each other.

```
# Applying crosstab for Weather and Season to obtain the contingency table.
contingency_table = pd.crosstab(df['weather'], df['season'])
```

```
values = chi2_contingency(contingency_table)
print(values)
```

```
Chi2ContingencyResult(statistic=np.float64(49.15865559689363), pvalue=np.float64(1.5499250736864862e-07), dof=9, expected_fr
       [6.99258130e+02, 7.11493845e+02, 7.11493845e+02, 7.11754180e+02],
       [2.11948742e+02, 2.15657450e+02, 2.15657450e+02, 2.15736359e+02],
       [2.46738931e-01, 2.51056403e-01, 2.51056403e-01, 2.51148264e-01]]))
```

```
if p_value < alpha:
  print("Reject the null hypothesis. Weather conditions and seasons are dependent on each other.")
else:
  print("Fail to reject the null hypothesis. Weather conditions and seasons are independent on each other")
```

```
Reject the null hypothesis. Weather conditions and seasons are dependent on each other.
```

Inference:

- Since the p-value obtained from the ChiSquare test of independence test is less than the significance level (α = 0.05), we reject the null hypothesis.
- This indicates that the weather conditions and seasons are dependent on each other.

Insights:

1. Working Day vs Demand

- Statistical testing indicates no significant difference in bike rental demand between working days and non-working days.
- This suggests that bike usage is not limited to work-related commuting and is also driven by leisure and personal travel.

2. Weather vs Demand

- One-way ANOVA results show that bike rental demand varies significantly across weather conditions.
- Clear and mildly cloudy weather conditions are associated with higher demand, while adverse weather conditions (rain, snow, storms) lead to a noticeable drop in rentals.

3. Season vs Demand

- Seasonal analysis reveals that bike rental demand differs significantly across seasons.
- Higher demand is observed during summer and fall, while winter and spring show comparatively lower demand, indicating seasonality in usage patterns.

4. Weather vs Season Relationship

- The Chi-Square test confirms that weather conditions are dependent on seasons.
- Certain weather types are more prevalent in specific seasons, which indirectly influences bike rental demand patterns.

5. User Contribution Insight (from correlation analysis)

- Registered users contribute more consistently to total demand, while casual users show spikes during favorable weather and seasons.
- Environmental factors primarily affect casual usage, whereas registered usage is relatively stable.

Business Recommendations for Yulu:

1. Weather-Aware Operations

- Use short term weather forecasts to optimize fleet deployment.
- Increase bike availability during clear and mildly cloudy days, and reduce operational exposure during adverse weather.

2. Seasonal Demand Planning

- Allocate more resources, maintenance capacity, and bikes during high demand seasons (summer and fall).
- Plan cost-optimization strategies during low demand seasons such as winter.

3. Focus on Registered Users

- Since registered users drive stable demand, Yulu should:
    - Strengthen subscription and loyalty programs
    - Offer seasonal or weather-based incentives to retain users year-round

4. Casual User Targeting

- Launch weather and season specific promotions (first ride discounts, referrals, weekend offers, New joiners offers) to convert casual users into frequent or registered users.

5. Data-driven readiness

- Continuous monitoring of seasonal and weather trends can help Yulu proactively manage supply and reduce revenue dips.

6. Data-Driven Expansion Strategy

- When expanding Yulu zones, prioritize areas where favorable seasonal and weather conditions align with high usage (e.g., office hubs during summer/fall).
- Avoid over-expansion in regions where adverse weather dominates most of the year.

```
Start coding or generate with AI.
```

```
Start coding or generate with AI.
```